# GW2025-HW9

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November 30, 2025

## Afterglow fitting

To fit to the observation, a model including the emission from a gaussian-jet (early afterglow) and a engine-fed kilonova is applied, according to Yang et al. (2022). The total emission from the kilonova is composed of two components: One comes from the r-process experienced by the mergers and the other is from the post-merger magnetar engine.

For the first component, a time-dependent expression for the heating rate (per ejection mass) of r-process is applied (Metzger et al., 2010; Metzger, 2019):

$$\dot{\varepsilon}_{\rm r} \sim 2 \times 10^{10} \epsilon (\frac{t}{1\,\rm day})^{-1.3}\,{\rm erg\,s^{-1}\,g^{-1}}\,, \tag{1}$$

where $\epsilon$ denotes the thermalization efficiency. Here the value is set as 0.5, and the radio active mass ratio $f_{\rm n}$ is 0.8 with $M_{\rm ej} = 0.037 M_{\rm sun}$.

As to the emission from the post-merger magnetar, the initial spin-down luminosity ($L_{\rm sd}$) is calculated with Eq (7) from Yang et al. (2022). Moreover, accounting for the decay of the luminosity, the magnetic dipole radiation formula is applied (Yu et al., 2013):

$$L_{\rm sd} = L_{\rm sd,0}(1 + \frac{t}{t_{sd}})^{-2}\,. \tag{2}$$

Here $t_{\rm sd}$ is the spin-down timescale, with the moment of inertia ($I_0$) set as $10^{45}\,{\rm g\,cm^2}$. The total ejection efficiency is written as 0.5*$0.03 e^{-1/\tau}$, where $\tau = 3\kappa M_{\rm ej}/(4\pi v_{\rm ej}^2 t^2)$.

With all the equations above, the afterglow emission + kilonova (black body) component is calculated, and the results are shown in Figure 1. It is found that the model can explain the observation well with best-fit parameters from Yang et al. (2022). It reveals the existence of a potential kilonova component from GRB 211211A.

## References

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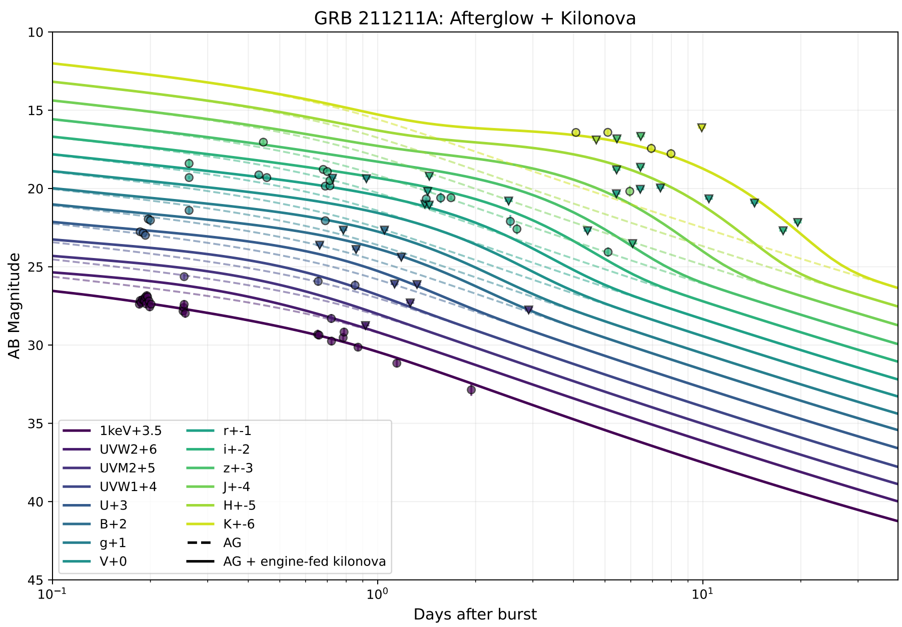


Figure 1: The fitting results for multi-wavelength observations, where the detections and upper limits are denoted by circles and nabla icons, individually. The dashed lines represent the afterglow emission and the solid lines show the total contribution of both afterglow and the engine-fed kilonova.